



CANDIDATE PREPARATION MATERIAL

Opportunity & Project Context Guide

Data Scientist

Prepared by Applaudo

Candidate Preparation Material

1. Welcome to the Opportunity

This guide is here to help you understand the opportunity in front of you — the role, the environment you'd be joining, and what success genuinely looks like — before your next conversations. Our goal is to make sure you're walking in with real context, not just a job title.

2. About Applaudo

Applaudo is a digital product and engineering company that partners with organizations to design, build, and scale technology — from data platforms to full product engineering. We work alongside our clients' teams as embedded senior talent, which means our engineers and scientists are expected to operate with real ownership from day one, not simply execute isolated tickets.

3. Project DNA

This engagement is with Bain & Company, a leading global management consulting firm, on a research and applied-ML initiative focused on business-entity matching. A few things shape the nature of the work:

- Real scale: the problem involves a unified business-entity graph built from 15+ third-party data sources, at a scale of hundreds of millions of entities — large enough that naive, brute-force approaches simply don't work.
- A production baseline already exists: there's an existing matching engine in production today, which serves as the comparison point for any new approach.
- Research cadence, not fixed sprints: the work moves in research and evaluation cycles rather than standard feature-delivery sprints — the deliverable is well-documented, defensible reasoning.
- A clean split between research and engineering: this role focuses on building and evaluating approaches; a separate, client-side engineering team is responsible for productionizing whatever gets validated.
- Genuinely messy data: names, aliases, domains, websites, firmographic attributes, multilingual records, and inconsistent hierarchies are all part of the everyday problem, not edge cases.

4. The Opportunity

At a glance

Role: Data Scientist

Client: Bain & Company

Engagement: 6 months

Working model: Remote, collaborating asynchronously across distributed time zones

Team: Part of a dedicated internal data-and-matching initiative, working alongside client-side engineering partners who own productionization

At its core, this role exists to figure out what actually works for matching business entities at massive scale — building and rigorously evaluating ML, embedding, and LLM-based approaches against a real production baseline, and turning ambiguous matching problems into clear, evidence-backed recommendations.

You would join as a senior, highly autonomous contributor. You set up your own experiments, define your own evaluation apparatus, and are trusted to report what you find — including when an approach doesn't work.

5. What You Would Work On

Rather than a list of isolated tasks, think of the work in four connected threads:

Building and evaluating matching approaches

Developing ML, embedding, and LLM-based approaches to matching company records, and designing scoring and ranking methods that separate true matches from duplicates, lookalikes, and unrelated entities.

Taming genuinely messy data

Working with name variations, aliases, domains, websites, firmographic attributes, multilingual records, and data hierarchies — where handling the mess well is itself a core part of the technical challenge.

Building the evaluation apparatus

Defining benchmark sets, metrics, baselines, and error-analysis processes so that any claim of improvement is genuinely defensible — and using LLM-assisted review both as a technique and as a quality ceiling to measure cheaper approaches against.

Communicating findings and recommendations

Turning ambiguous matching problems into explicit hypotheses, experiments, and recommendations, and explaining model behavior and trade-offs clearly to both engineering and business partners — including reporting negative results as clearly as positive ones.

Productionizing accepted approaches is explicitly out of scope for this role — that responsibility sits with the client's engineering partners. Your accountability is for the rigor and honesty of the experiment, not for shipping it to production.

6. Technology & Technical Environment

Priority	Topics
Core	• Python and SQL for large-scale data work • Embeddings and semantic similarity • Practical, cost-aware application of LLMs • ML frameworks: TensorFlow, PyTorch, PyCaret, or equivalent
Supporting	• Classification and NLP techniques

Priority	Topics
	• Neural network and transformer architecture fundamentals • Experimental design and evaluation tooling
Nice to have	• Entity resolution, record linkage, or deduplication experience • Ranking, similarity scoring, retrieval, or clustering techniques • Spark, Snowflake, Databricks, or BigQuery • Multilingual dataset experience • Company, domain, website, or firmographic data experience

7. What the Team Is Really Looking For

Beyond the technology list, a few underlying signals matter most in how candidates are evaluated for this role:

Intellectual honesty

A willingness to report a disproved hypothesis as clearly and confidently as a successful one — the engagement depends on trustworthy findings, not just positive ones.

Genuine autonomy

The ability to set up your own experimental pipeline and research approach without waiting for detailed direction.

Cost- and scale-aware design

Reasoning about inference economics and scalability from the outset, at a scale where naive approaches simply break down.

Rigorous experimental judgment

Defining baselines, metrics, and test sets that let you say with confidence whether quality actually improved — not just reporting a single accuracy number.

Clear, resilient communication

The ability to explain trade-offs and defend a recommendation to engineering and business partners who will challenge it, in writing and in conversation.

8. What Success Could Look Like

First weeks

- Building a working understanding of the existing production matching engine and the available data sources.
- Standing up your own experimental pipeline and initial benchmark set.

First months

- Delivering a defined evaluation framework and the first candidate matching approaches with a clear, measured comparison against the existing baseline.
- Producing well-documented experiments, including any that disproves an initial hypothesis.

Longer-term contribution

- A body of rigorous, well-documented research that shapes which matching approaches the client's engineering team ultimately builds into production.
- Being a trusted, autonomous research partner, whose recommendations are taken seriously precisely because your reasoning holds up under scrutiny.

9. How to Position Your Experience

As you prepare, think in terms of what you can authentically stand behind under detailed follow-up questions. Be ready to discuss:

- An ML approach you built and evaluated end-to-end, from hypothesis to result.
- A production classification model you retrained or maintained — not notebook-only work.
- A messy or multilingual data problem you solved.
- A cost or scale trade-off you navigated deliberately.
- An experiment that disproved your hypothesis, and how you reported it.
- A recommendation you defended to a stakeholder who pushed back on it.

Precision matters more than breadth here. One example you can go three questions deep on is worth more than five you can only summarize at a high level.

10. What This Role Is / What This Role Is Not

This role IS	This role IS NOT
✓ Research- and experimentation-focused Data Science	✗ Productionizing accepted approaches
✓ Hands-on ML, embedding, and LLM-based matching work	✗ Owning the matching platform's architecture, pipelines, or infrastructure
✓ Autonomous experimental design and evaluation	✗ Manual match adjudication at volume
✓ Clear, honest technical communication, including negative results	✗ Product management or requirements gathering

11. Interview Journey

At a high level, the selection process involves two stages, run by two different parties:

Applaudo evaluation

An internal conversation with Applaudo's technical leadership to assess depth, communication, and fit for the role — this is what the companion Technical Interview Study Guide is designed to help you prepare for.

Client evaluation

The client's process typically includes one to two conversations: an initial discussion with leadership, followed by a technical conversation with members of the project team.

We'll walk you through the specifics of format and timing for each stage as they're scheduled.

12. Final Note

We want you to walk into these conversations with a clear, honest picture of what this opportunity involves, so you can show us your strongest, most authentic experience. If anything in this guide raises questions, bring them to your recruiter; we're glad to talk it through.